

NSDC Project Spring 2025 Week 3

Introductions and Deciding Project Idea (Meeting 4/17)

Introductions:

- Name
- Major
- Year
- What you know about Machine learning
- What they want to learn or are most excited about for this project

Michelle Ly:

3rd year cog sci

CNNs, ML not too much experience

Harry Kong:

3rd year stats and DS

Not much abt ML only from COMM 188C

Dylan Aubrey:

2nd year stats DS

Not much abt ML

Has been in an NSDC project but more for data visualization

Michelle Lin:

1st year stats DS

1st NSDC project

Project Expectations

Six-week long project where we will build our own machine learning algorithms from scratch using Python, and then implement these algorithms on a dataset on a topic of our choice. We'll also create a slideshow presentation and present what we've learned for the project showcase

Project Showcase Date: **May 29th, 2025**

We will learn about the following machine learning algorithms:

- 1) Linear Regression [EASY]
- 2) K-Nearest Neighbors Regression [EASY]
- 3) Logistic Regression [EASY]
- 4) K-Nearest Neighbors [EASY]
- 5) Naive Bayes [EASY]

- 6) Decision Tree Regression [MEDIUM]
- 7) Decision Tree Classifier [MEDIUM]
- 8) Support Vector Machine [MEDIUM-HARD]
- 9) Gradient Boosting Regressor [HARD]
- 10) Random Forest [HARD]
- 11) Neural Networks [VERY HARD]

Divide the project members into groups of two and each group will work on learning two Machine Learning Algorithms:

- **Group 1 (Linear Regression, Decision Tree Regression, Decision Tree Classifier):**
Ryan So, Sophia Knapp
- **Group 2 (K-Nearest Neighbors Regression, K-Nearest Neighbors, Gradient Boosting Regressor):** Harry Kong, Michelle Ly
- **Group 3 (Logistic Regression, Support Vector Machine, Random Forest):** Michelle Lin, Bernice Wong, Dylan Aubrey
- **Group 4 (Naive Bayes, Neural Networks):** Kyle He, Daniel Perez

Any Project Topic Ideas?

Let's decide on a idea and look at datasets for what we want to utilize our machine learning models later on in the project

Datasets do not have to be finalized yet, but the topic needs to be finalized

Search on [kaggle.com](https://www.kaggle.com) for good datasets if no ideas are given

Ideas:

- Environmental Impact
 - <https://www.kaggle.com/datasets/adityakadiwal/water-potability>
 - Classify water potability using logistic regression, Random Forest, or XGBoost.
 -
- Pokemon Battling - <https://www.kaggle.com/datasets/alopez247/pokemon>

Finalized Idea:

Pokemon Battling

Action Items for Next Week:

- Get in assigned pairs and learn about your chosen algorithms conceptually
- Prepare to teach the rest of us how your chosen algorithms work

NSDC Project Spring 2025 Week 4

Learn About Machine Learning Algorithms (Meeting 4/23)

Groups:

- **Group 1 (Linear Regression, Decision Tree Regression, Decision Tree Classifier):**
Ryan So, Sophia Knapp
- **Group 2 (K-Nearest Neighbors Regression, K-Nearest Neighbors, Gradient Boosting Regressor):** Harry Kong, Michelle Ly
- **Group 3 (Logistic Regression, Support Vector Machine, Random Forest):** Michelle Lin, Bernice Wong, Dylan Aubrey
- **Group 4 (Naive Bayes, Neural Networks):** Kyle He, Daniel Perez

Machine Learning Algorithms:

- **Random Forest**
- **Logistic Regression**
- **Support Vector Machine**
- **Linear Regression**
- **Decision Tree Regression**
- **Decision Tree Classifier**
- **K-Nearest Neighbors Regression**
- **K-Nearest Neighbors**
- **Gradient Boosting Regressor**
- **Naive Bayes**
- **Neural Networks**

Decide What We want to Predict Using our Dataset:

<https://www.kaggle.com/datasets/alopez247/pokemon?resource=download>

Ideas:

- **Catch rate (What we decided to predict)**
- **Pokemon ID in the Pokedex**
- **If it is legendary**
- **Predicting pokemon types**

Separate into Groups for Dataset Exploration and Preprocessing and ML Algorithm

Implementation:

- **Dataset Exploration/Preprocessing Team:** Harry Kong, Sophia Knapp, Daniel Perez, Kyle He
 - Explore dataset, create visualizations
 - Clean data or possibly perform feature engineering

- **ML Algorithm Implementation Team:** Michelle Ly, Michelle Lin, Bernice Wong, Dylan Aubrey, Ryan So
 - Implement Regression models since we will be trying to predict the catch rate
 - This will include:
 - Linear regression
 - K-Nearest Neighbors Regression
 - Decision Tree regression
 - Gradient Boosting Regressor
 - Random Forest Regressor

Action Items for Next Week:

- Start exploring and cleaning the dataset, performing any visualizations that would be cool as well.
- Start on implementing the machine learning algorithms we will need for this project

NSDC Project Spring 2025 Week 5

Check-in with Each Team (Meeting 5/2)

Project Showcase has been decided: **May 27th 7-9pm!**

Let me know if that time is an issue for you

Dataset Exploration/Preprocessing Team Updates:

Is any data cleaning or encoding necessary?

Anything interesting about the dataset?

Begun creating visualizations?

Expect one visualization from each person, vary the visualizations, simpler is better

ML Algorithm Implementation Team:

Implement the algorithms using only NumPy, without using scikit-learn or any other prepackaged algorithms. Implement the algorithms so that it can be applied to any dataset.

Structure:

Define a class and put necessary functions within that class.

For example, for Linear Regression,

class LinearRegression:

 define initializing variables

 define any necessary functions for the algorithm

Divide up the algorithms so each person has one to implement. One person will have to implement two algorithms.

- Linear Regression (Ryan So)
- K-Nearest Neighbors Regression (Michelle Lin)
- Decision Tree Regression (Dylan Aubrey)
- Gradient Boosting Regressor (Michelle Ly)
- Random Forest Regressor (Bernice Wong)

Action Items for Next Week:

- Dataset Exploration/Preprocessing Team:
 - Each create one visualization for next week
- ML Algorithm Implementation Team:
 - Implement the algorithms from scratch using only NumPy

NSDC Project Spring 2025 Week 6

Visualization Fine-tuning and Beginning Model Development (5/8)

Important: Mandatory Professor Panel Next Week Monday 5/12 at 7pm in Haines A25.

All project leads and project groups will be there so it's also a good opportunity to meet other groups and hear what other groups have been up to!

Admission is free, and it's a good opportunity to hear about our own UCLA professors' research in all different types of fields in Data Science. It can provide inspiration for what you want to do in your own career in the future or provide ideas.

Here is the Professor Panel:

https://docs.google.com/document/d/1QK9q01HYw3VYLE_PbbbXZKZ7RgPXo7exwLm04_AeIRU/edit?tab=t.0

It would be great if you guys prepared a couple questions for the professors as well

Dataset Exploration/Preprocessing Team Updates:

Let's take a look at the visualizations

Visualizations:

Good work just continue to work on the visualizations and keep in mind to keep the labels clear and use words

Fine-tune the visualizations and also prepare some visualizations regarding model performance when the ML Algorithm Implementation Team is finished training the models

ML Algorithm Implementation Team:

Report your progress

- Linear Regression (Ryan So)
- K-Nearest Neighbors Regression (Michelle Lin)
- Decision Tree Regression (Dylan Aubrey)
- Gradient Boosting Regressor (Michelle Ly)
- Random Forest Regressor (Bernice Wong)

Start training the implemented algorithms using NumPy only

Steps for training a machine learning model:

- 1) Split the data into a training set and testing set (can be done with only NumPy)
- 2) Fit each implemented algorithm with the training set (easier to do it all at once with a loop)
- 3) Evaluate the models using MSE and R squared since they are Regression models

Action Items for Next Week:

- Create Separate channels for Dataset Exploration/Preprocessing Team and ML Algorithm Implementation Team
- Dataset Exploration/Preprocessing Team:
 - Fine-tune the visualizations, can also visualize the model performance to see which model works the best
- ML Algorithm Implementation Team:
 - Move on to model training and evaluation still using only NumPy

NSDC Project Spring 2025 Week 7

Work on Improving Model Performance and Telling a Story (5/16)

Dataset Exploration/Preprocessing Team Updates:

Update progress on our visualizations

Start making final visualizations and visualizing which models are the best performing ones

Remember to tell a story through visualization. The visualization should have a purpose, we're not just making good looking visualizations for fun

Write one short paragraph with the final visualization you choose, analyzing its importance in the context of our data

In our future careers, as data scientists we aren't expected just to create models and visualizations but also explain why it's important to a certain company, etc.

ML Algorithm Implementation Team:

All the algorithms are implemented, let's look at how to train each model now on our dataset

We will evaluate our Regression models on R squared and Mean Squared Error:

- R squared represents the proportion of variance in the dependent variable that can be explained by our independent variables; we want this value to be as close to 1 as possible
- Mean Absolute Error is the average distance between predicted and actual values and tells us about the accuracy of a model's predictions; we want this value to be as close to 0 as possible

Also work on improving model performance through hyperparameter tuning

Hyperparameters are parameters that are not learned through training but are set beforehand

Examples would include learning rate, batch size, even the percentage of training to testing data

Couple ways to perform hyperparameter tuning (There are more methods too but we'll focus on these):

1. Grid Search (Computationally expensive)
2. Random Search (Takes random combinations of hyperparameters)

If time permits, compare the model performance for the models we implemented ourselves against sklearn's models

WORK TIME

Action Items for Next Week:

- Train models on our implemented algorithms
- Use hyperparameter tuning to improve our model
- Finalize the best visualization and write one paragraph analyzing its importance in the context of this project
- Begin working on our slideshow:
https://docs.google.com/presentation/d/1xSqxfpuQccA_BuNOm8Ag_-oymPX8O6r5OJuB34Lu1e8/edit?slide=id.p#slide=id.p (TWO FINAL VISUALIZATIONS FROM DATA PREPROCESSING/VISUALIZATION TEAM & TWO SLIDES EACH FOR EACH MACHINE LEARNING ALGORITHM)

NSDC Project Spring 2025 Week 8

Finishing Slideshow and Presentation (5/23)

Project Showcase Announcements

Project showcase is the **Tuesday of Week 9, 5/27 at 7pm in the Broad Arts Center 2100A**. We will be presenting our project!

Each group will be allowed 10 minutes maximum amount of time to present.

I will try to get us to present within 8-8:30pm time frame, since there are people that can only make this time

Showcase Slideshow

https://docs.google.com/presentation/d/1xSqxfpuQccA_BuNOm8Ag_-oymPX8O6r5OJuB34Lu1e8/edit?slide=id.g35888ab3e0e_4_0#slide=id.g35888ab3e0e_4_0

Data Preprocessing/Visualization Team

Choose the 2 best visualizations to present

Briefly explain any encoding done, any data preprocessing we performed for our models, and also feature selection during the presentation

Remember to explain why the visualization is important or why we explored what we explored about the dataset in relation to machine learning

So what is the distribution of the data, what are these visualizations telling us about what the relationships between our independent variables and our dependent variable, catch rate

Machine Learning Algorithm Implementation Team

All models should be finished training and tuned.

Have 2 slides each

1st slide:

- How the algorithm works
- How it was implemented
- How the model was trained and how was it improved
- Comparison between the sklearn model and homemade model

2nd slide:

- Homemade model performance based on R squared and MAE
- Sklearn model performance based on R squared and MAE

- What differences are there and why might that be so?

Work Time

Work on the slideshow, needs to be finished by TONIGHT

Action Items for Next Week:

- Finish the slides by tonight
- Be ready for project showcase on Tuesday
- Will send out more details about the presentation over the weekend

NSDC Project Spring 2025 Week 9

Project Showcase 7-9pm at Broad Arts Center 2100A (5/27)